



BLE and surveillance camera-based navigation system for visually impaired people

Sok-Ian Sou¹ · Syuan-Min Huang¹

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Abstract

This study introduces ViTrack, an advanced vision-and-wireless-based navigation system that assists visually impaired individuals in indoor environments in a user-controlled, privacy-preserving manner. Traditional visual navigation methods often encounter challenges due to obstructed views, hindering effective tracking and identification. Similarly, wireless positioning systems are prone to signal interference, leading to inaccuracies. ViTrack seamlessly blends vision-based and wireless positioning technologies to provide comprehensive, unobtrusive navigation support. A key element of our approach is using existing surveillance cameras, expanding our observation capabilities while minimizing the need for additional devices for the user. Our system is designed to improve navigation while maintaining simplicity for the user. We exhibit ViTrack within an already-established surveillance camera network. It automatically employs low-cost wireless sniffers to detect users within the observation guiding zone. Simultaneously, vision-based methods ensure collision-free path planning. Our findings highlight the superiority of Long Short-Term Memory (LSTM) neural networks over other neural network architectures regarding localization and tracking accuracy. We also demonstrate the crucial role of data augmentation in mitigating device failure impacts, significantly boosting the system's reliability.

Keywords Visually impaired individuals · Vision-based navigation system · Wireless fingerprinting · Deviation detection · Obstacle avoidance

✉ Sok-Ian Sou
sisou@mail.ncku.edu.tw

Syuan-Min Huang
n26100757@gs.ncku.edu.tw

¹ Department of Electrical Engineering, National Cheng Kung University, University Road 1, Tainan 701, Taiwan

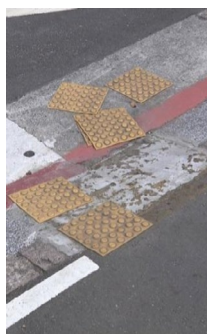
1 Introduction

According to the World Health Organization (WHO), 2.2 billion people globally have vision impairment, with approximately 1 billion cases either preventable or still requiring attention [1]. This underscores the urgent need for technological solutions that can significantly improve the quality of life for visually impaired individuals. While numerous studies have explored the quality of life among individuals with visual impairment, Bonsaksen et al. address this gap by evaluating quality of life in people with visual impairment relative to the general population and investigating the influence of adverse life events and psychosocial factors [2]. Recent studies have demonstrated that tactile paving can significantly enhance the street-crossing performance of people with visual impairment by helping them maintain a straight path, reduce directional deviation and crossing time, and improve gait regularity and symmetry [3]. However, the effectiveness of traditional navigation aids such as white canes and tactile pavings is often compromised by inconsistent implementation and poor maintenance. As illustrated in Fig. 1, obstructions severely impact the mobility and safety of individuals who rely on tactile navigation. Numerous official guidelines and research studies emphasize that static obstructions, such as improperly parked bicycles or vehicles, are among the most persistent and disruptive barriers to safe navigation for visually impaired individuals. For example, the UK Department for Transport's Guidance on the Use of Tactile Paving Surfaces [4] notes that blocked tactile paths critically undermine their intended function. Supporting this, a major RNIB survey [5] found that over 90% of blind and partially sighted people reported colliding with street obstacles in their local area over three months, with a third of these incidents resulting in injury. The most common obstacles encountered were parked cars (70%), bins (64%), fixed street furniture (60%), and advertising boards (49%), highlighting static obstructions as a persistent and hazardous barrier to independent mobility. These findings underscore the importance of identifying and addressing static obstacles to improve safety and accessibility for visually impaired individuals, particularly in urban and campus environments.

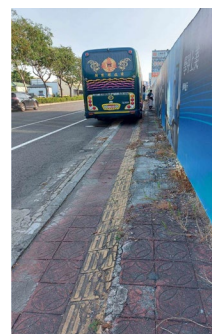
Recent surveys [6, 7] have critically assessed the landscape of assistive technologies, identifying both advancements and ongoing limitations. While wearable solutions such as the RGB camera system by Tyagi et al. [8] offer valuable functionality, they often present cost, convenience, and user acceptance challenges. Sensor-based Electronic Travel Aids (ETAs) using infrared [9],



(a) Obstruction



(b) Poor maintenance



(c) Inaccessible pathway

Fig. 1 Common issues affecting tactile pavings: (a) physical obstruction, (b) poor maintenance, and (c) pathway inaccessibility

ultrasound [10], RFID [11], GPS [12], BLE [13], and Wi-Fi [14] have contributed to localization and navigation, but many require users to manage extra devices or rely on specialized infrastructure. Vision-based ETAs, while providing detailed environmental understanding, often suffer from high computational demands and limited field of view. To address these challenges, we propose ViTrack, a vision-based platform that leverages existing surveillance camera networks and BLE/Wi-Fi fingerprinting localization to guide visually impaired individuals. ViTrack processes real-time video streams and wireless signal data via users' smartphones, providing audio-based navigation guidance without additional equipment. By adopting a third-person perspective, ViTrack offers a broader field of view for obstacle detection and navigation, overcoming the blind spot limitations of first-person systems and ensuring user privacy and convenience.

In [15], the authors provide a recent review of existing ETA systems. Table 1 provides a comparative overview of ViTrack alongside selected ETA solutions popular among visually impaired users. vOICe [16] represents an advanced wearable system converting visual data into auditory soundscapes, but involves significant hardware cost and user adaptation. Microsoft Soundscape [17] and BlindSquare [18] are smartphone-based navigation applications leveraging GPS and audio cues, with Soundscape focused mostly on outdoor spatial awareness and BlindSquare extending support to indoor environments. Unlike these approaches, ViTrack relies on BLE signal fusion and third-person infrastructure cameras to deliver robust indoor navigation without burdening users with additional devices. This infrastructure-centric modality offers fine-grained localization and obstacle detection in indoor spaces, addressing a notable deficiency in many existing ETAs. Additionally, ViTrack's lightweight and user-controlled design ensures low-cost deployment and user comfort. This comparison underscores ViTrack's potential to bridge current gaps by combining technological innovation, economic accessibility, and user-centered design tailored for indoor assistive navigation.

The ViTrack system offers a cost-effective and non-intrusive solution by leveraging existing cameras and user smartphones, in contrast to traditional wearable devices. This innovative approach not only promotes sustainability but also encourages wider adoption. While the current study focuses on single-user scenarios and detecting static obstacles, the architecture is designed to scale for multiple users and more complex environments. The initial implementation was carried out in a controlled setting with limited participants; hence, real-world deployment will require further evaluation with a broader and more diverse group in dynamic environments. The main contributions of this study are as follows:

- Overcoming navigation blind spots through a third-person perspective approach utilizing surveillance technology.
- Addressing the challenge of person identification by presenting a user matching algorithm that integrates wireless signals and visual images.
- Proposing deviation detection and obstacle avoidance algorithms for route planning in virtual paving to assist visually impaired individuals in walking safely.

The remainder of this paper is organized as follows. Section 2 introduces the ViTrack architecture and its application in the National Cheng Kung University (NCKU) case study. Section 3 discusses the wireless-based user matching approach, highlighting the integration of wireless signal localization and deep learning with visual tracking using Particle Filters and YOLOv7. Section 4 outlines the visual-based guiding system, focusing on human orientation estimation, deviation detection, and obstacle avoidance algorithms. Section 5

Table 1 Comparison of ViTrack and selected ETA systems

Feature	ViTrack	vOICe	Microsoft Soundscape	BlindSquare
Cost	\\$0 (smart-phone app)	Approx. \\$1000 (hardware + software)	Free (software only)	Paid app (\\$50)
Hardware	Smart-phone + infrastructure	Head-mounted camera + headphones	Smartphone	Smartphone
Power	Phone battery	Separate device battery	Phone battery	Phone battery
Comfort	Familiar device	Wearable, can be intrusive	Familiar phone	Familiar phone
Sensing Modality	BLE + infrastructure camera fusion	Real-time image-to-sound translation	GPS + audio cues using 3D audio	GPS + local databases for indoor/outdoor
Navigation Support	Indoor optimized with multi-sensor fusion	Outdoor/ indoor position via soundscape	Primarily outdoor navigation	Indoor and outdoor navigation
Feedback	Audio + vibration through phone	Auditory soundscapes	3D spatial audio cues	Audio cues, voice guidance
Setup Complexity	Minimal app install	Device calibration + training needed	App install	App install
User Training	Minimal	Requires significant training	Minimal	Minimal
Privacy	User-controlled, anonymized	Device captures camera view	User-controlled	User-controlled

details the experimental setup, including test locations, devices, sniffer placement, route definitions, and data collection. Section 6 assesses system performance, highlighting localization accuracy, sniffer failure robustness, and body orientation precision. Finally, Section 7 concludes the paper and discusses future research directions.

2 Video surveillance system for smart campus

In recent years, the issue of campus safety has gained paramount importance; governments and educational institutions have made substantial investments in deploying campus surveillance cameras, considering them a cornerstone in their prevention strategies. This section offers an illustrative real case study of a campus safety network based on the surveillance system installation at NCKU campus in Taiwan (Fig. 2).

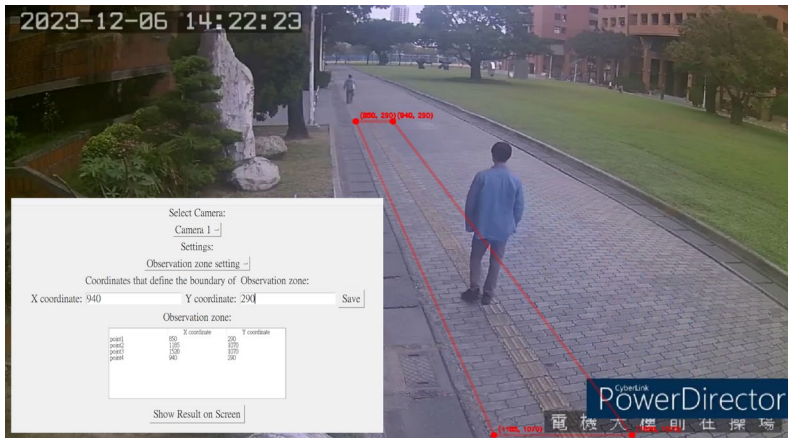


Fig. 2 Example of configuring the Observation Zone in the ViTrack system. The main image shows the defined zone boundaries (red lines) on the campus pathway, while the inset illustrates the user interface to configure the zone coordinates

Figure 3 depicts the ViTrack system interacting with several essential components. The Campus Safety Network (Fig. 3 (a)) works over an independent intranet with more than 500 surveillance cameras and 69 network video recorder (NVR) servers. Each camera produces video transmissions with a resolution of 1920×1080 pixels at a frame rate of 20 frames per second. Based on the framework in [19], the system also employs MapTalk (see Fig. 3 (c)) to provide a web interface that can communicate with the YOLO device (Fig. 3 (b)) to manipulate cameras easily by selecting the camera icon and showing the real-time monitor screen of the corresponding camera. It also safely provides real-time camera streams to other innovative applications,

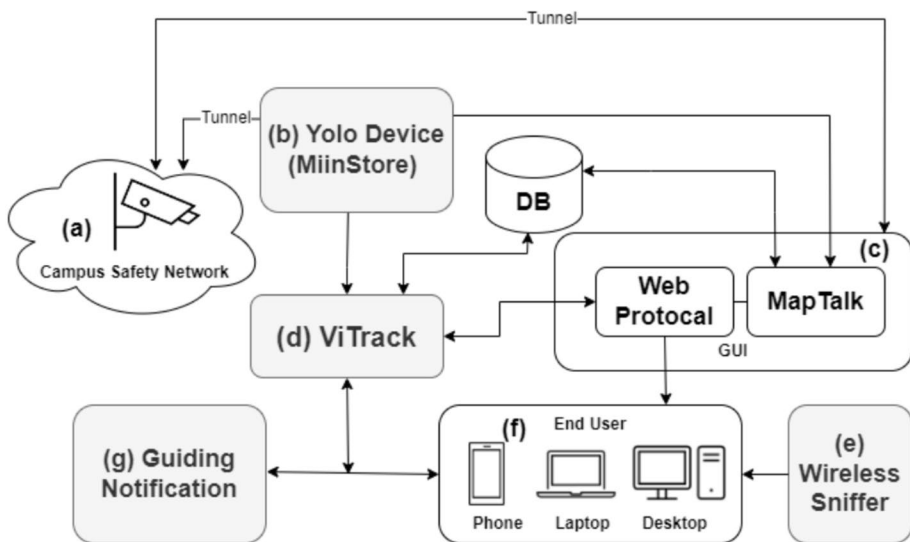


Fig. 3 System architecture of the NCKU ViTrack platform. Data flows and connections between components highlight the end-to-end process from sensor input to user guidance

such as ViTrack (see Fig. 3 (d)). Here, we access camera streams from the monitors within the intranet and facilitate the establishment of Secure Shell (SSH) tunnels between the monitors and ViTrack. Our proposed ViTrack mainly works over the MapTalk and the YOLO device.

In Fig. 3, the end user (a.k.a. ViTrack administration) (Fig. 3 (f)) can interact with ViTrack through the web protocol for system configuration and user record maintenance stored in the database (DB). The ViTrack end user receives the wireless sniffer data (i.e., the Received Signal Strength Indicator (RSSI) wireless signal data) through the deployed sniffers nearby. Once the end user receives ViTrack sniffer signals, it can send a “Tracking Request” to ViTrack and receive a “Guiding Notification” on mobile devices. Once the ViTrack system receives the “Tracking Request” from the end user, it will communicate with the YOLO device, which connects with the campus safety network. Details of the design of the ViTrack system will be given later.

In the MapTalk system, various network objects, such as cameras, are conveniently listed in a dropdown menu, as highlighted by the blue rectangle in Fig. 4 (1). Users can interact with this menu to select one or multiple network objects, like cameras, for display. These selected objects are then visually represented on the map according to their actual physical locations using icons. In cases where multiple cameras are located nearby, a single icon represents them collectively to ensure map clarity. The status of these cameras is easily discernible through color-coded icons: a blue icon, shown in the black rectangle in Fig. 4 (2), indicates all cameras are operational, while a purple icon, as seen in the red rectangle in Fig. 4 (3), signifies that all cameras are non-functional. Furthermore, an icon that is half blue and half purple, depicted in the orange rectangle in Fig. 4 (4), means some cameras at that location are functional, and others are not. MapTalk also provides the flexibility to integrate new cameras via the administration webpage. When a user clicks on any camera icon on the map, the monitoring screen for that specific camera is displayed, as demonstrated in Fig. 4 (5).

Figure 5 illustrates the strategic positioning of surveillance cameras across the Tzu-Chang Campus at NCKU. As indicated in Fig. 5a, many surveillance cameras are installed outdoors to monitor public spaces and enhance campus security. These cameras are predominantly situated at the main building entrances across the campus. This study aims to leverage this extensive network of pre-existing cameras to enhance navigational assistance for visually impaired individuals and reinforce overall campus safety measures. Additionally, Fig. 5b provides a bird’s-eye view with satellite imagery from Google Earth, marking camera locations with yellow pins to represent their distribution visually. The orange arrow indicates the site chosen for in-depth experimental analysis, providing insight into the focal point of interest.

A detailed view of one such experimental camera is presented in Fig. 5c, highlighting its mounting and the area it oversees. Positioned on a metal pillar outside the Department of Electrical Engineering, this camera is selected for its strategic vantage point. It covers a significant stretch of 16 meters of tactile paving that pedestrians frequent. This placement is critical for our experiment, aiming to demonstrate the system’s effectiveness in a busy, real-world setting.

ViTrack leverages the existing camera infrastructure at NCKU, eliminating the need for hardware upgrades. It employs YOLOv7 for real-time object detection, which was selected for its efficiency and robust performance. The system’s modular design facilitates the integration of alternative models if necessary. ViTrack can coordinate multiple cameras through person re-identification techniques, and preliminary tests have demonstrated its feasibility. The compatibility of YOLOv7 with the platform and its precision in real-time applications makes it an ideal choice. Furthermore, ViTrack does not necessitate separate saliency detection, as YOLOv7’s architecture inherently highlights critical regions, thus avoiding additional latency.

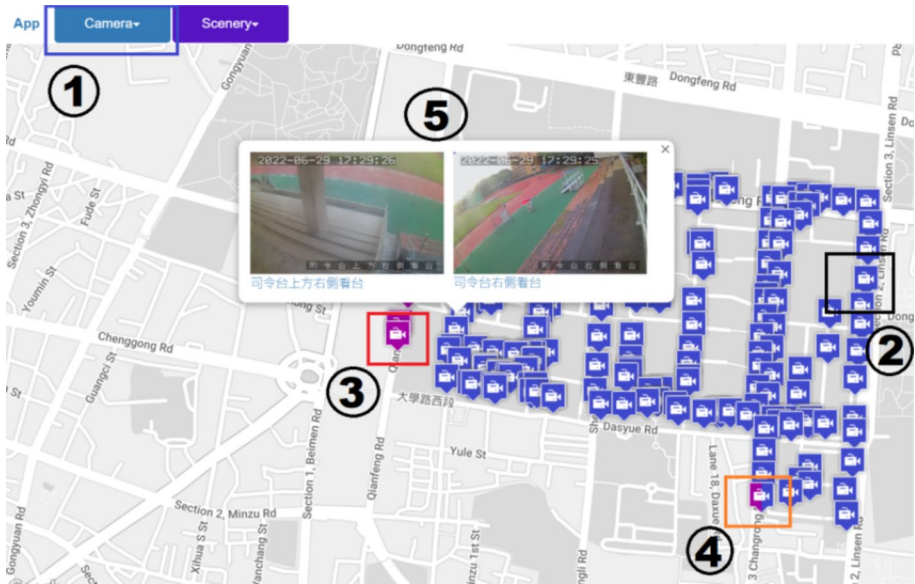
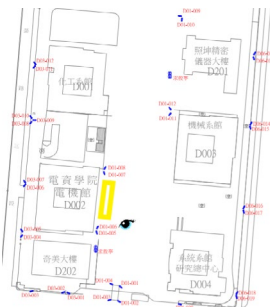


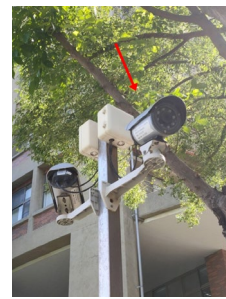
Fig. 4 Video management interface of the ViTrack system using MapTalk in NCKU. The map displays the locations of surveillance cameras across campus. This interface enables efficient monitoring and management of video streams for user guidance and safety



(a) Camera deployment



(b) Aerial Overview



(c) On-Site Camera Setup

Fig. 5 Deployment of the ViTrack surveillance system at NCKU (Tzu-Chang Campus)

3 Wireless-based user matching

Figure 6 illustrates the functional blocks of ViTrack, encompassing the user matching and guiding phases. The user matching phase involves deploying several sniffers to assist in localization and initiate guidance assistance. In the guiding phase, data captured by the camera is utilized to detect and guide visually impaired individuals on the appropriate path.

This section focuses on using wireless signals to enable the matching of the target ViTrack visually impaired user in camera-covered areas. Initially, low-cost wireless sniffers are positioned within the Observation Zone (OZ) to detect users entering the area and collect RSSI values. In conjunction with a supervised learning model trained on historical data,

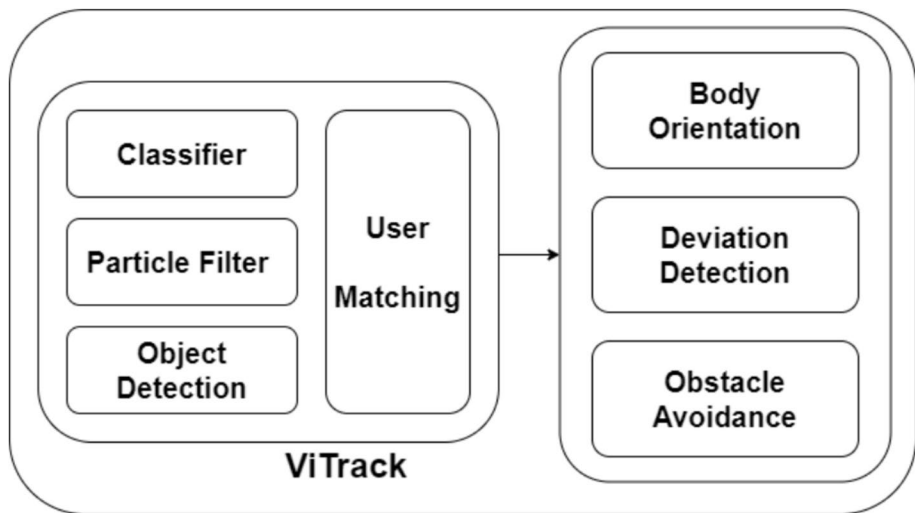


Fig. 6 Block diagram of the ViTrack system's user matching and guidance notification modules

these values assist ViTrack in precisely determining the user's location within the OZ. Once identified, ViTrack incorporates visual techniques for user tracking and navigation.

The system incorporates a BLE interface for automatic detection to facilitate ease of use, which is especially effective near building entrances where camera surveillance is concentrated. This proximity-based approach simplifies access to services for visually impaired users by reducing manual smartphone operation. While voice activation could serve as an input for activating ViTrack, the system's functionality is limited to camera-covered areas, primarily around building entrances on campus.

This study defines a reference point (RP) as an OZ Block (OB). Sniffers deployed in the area capture BLE advertising messages and record device identity and RSSI values. These form the fingerprint data, represented as $[RSSI_1, RSSI_2, \dots, RSSI_n]$, by collating records from n sniffers for the same device. We use DNN models to train our location classifiers, learning to map these wireless features to specific OBs. Additionally, particle filters are employed to track user locations. The DNN model outputs probabilities for each OB; the OB with the highest probability is identified as the estimated location.

During the testing phase, our approach integrates visual and wireless signal data to identify users within the camera's view. Since the camera footage may include both relevant users and bystanders, we utilize BLE sniffer data to estimate the positions of the target users. A particle filter is then employed for refined position tracking. Once a user is correctly matched, the navigation process begins. This includes monitoring for any deviation from the designated path, alerting the user to avoid obstacles, and using environmental obstacle data to enhance the safety and accuracy of navigation.

For demonstration purposes, we employ a camera that monitors an area with tactile paving to illustrate its real-world deployment. However, considering that this is a virtual tactile paving in our experiment, the user cannot feel the special texture of the paving and will easily deviate from the path. This paper primarily addresses the problems of walking deviation and obstruction. It is worth mentioning that the system does not require any wearable devices and only requires the user's smartphone with Bluetooth functionality enabled to operate.

3.1 Model training

To address the variability in RSSI measurements arising from heterogeneous hardware, our system incorporates a min-max normalization step to ensure cross-device compatibility. Specifically, for each BLE sniffer j ($j \in \{1, 2, \dots, J\}$), we collect N RSSI measurements, which are organized into a vector $s_j \in \mathbb{R}^N$ as shown in (1):

$$s_j = [\text{RSSI}_{1,j}, \text{RSSI}_{2,j}, \dots, \text{RSSI}_{N,j}]. \quad (1)$$

To ensure that the signal strengths are comparable across all devices, each RSSI value is normalized to the interval $[0, 1]$ using min-max normalization, as defined in (2):

$$\text{RSSI}'_{i,j} = \frac{\text{RSSI}_{i,j} - \min(s_j)}{\max(s_j) - \min(s_j)}, \quad (2)$$

where $\text{RSSI}_{i,j}$ represents the i -th RSSI measurement from the j -th sniffer. The resulting normalized values from all sniffers are then concatenated to form a feature vector for each measurement instance, as expressed in (3):

$$r_i = [\text{RSSI}'_{i,1}, \text{RSSI}'_{i,2}, \dots, \text{RSSI}'_{i,J}]. \quad (3)$$

To further enhance the robustness of the localization model, particularly in scenarios where some sniffers may fail, we employ data augmentation. This is achieved by randomly selecting a sniffer index $a \in \{1, \dots, J\}$ and setting its corresponding normalized RSSI value to zero, thereby simulating a device outage. The augmented feature vector is given by (4):

$$r_{i,\text{aug}} = [\text{RSSI}'_{i,1}, \dots, \text{RSSI}'_{i,a-1}, 0, \text{RSSI}'_{i,a+1}, \dots, \text{RSSI}'_{i,J}]. \quad (4)$$

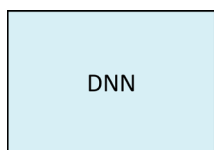
The original and augmented feature vectors are utilized during model training to improve the system's resilience to real-world hardware failures.

For the fingerprinting localization task, we investigate three neural network architectures: a deep neural network (DNN), a one-dimensional convolutional neural network (1D-CNN), and a long short-term memory (LSTM) network. The detailed parameter settings for each model are summarized in Table 2. In all models, the output layer employs the Softmax activation function, which is appropriate for the classification nature of the localization problem. Consequently, the number of neurons in the output layer is equal to the number of reference points (RPs), and the output vector represents the probability distribution over all possible RPs, indicating the likelihood of the user's location.

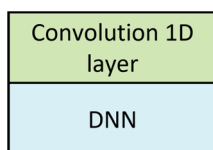
The DNN model serves as the baseline, while the 1D-CNN and LSTM architectures are constructed by introducing a convolutional layer and an LSTM layer, respectively, as illustrated in Fig. 7. Both the convolutional and LSTM layers are configured with 20 filters or units, and additional fully connected layers are included to ensure compatibility with the DNN input size. Furthermore, the 1D-CNN and LSTM models incorporate a temporal dimension called the time step T . In our experiments, we consider $T = 1$ and $T = 3$ to examine the impact of temporal information on localization accuracy. Notably, in the 1D-CNN model, the kernel size of the convolutional layer is set equal to the time step T .

Table 2 Parameters of deep learning models

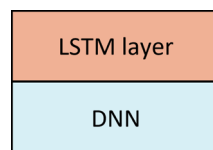
Parameter	DNN	1D-CNN	LSTM
Base architecture	2 hidden layers	1 conv. layer	1 LSTM layer
Layer specifics	-	Filters: 20 Kernel: 3, Stride: 1	Hidden size: 20
Dense layer (FC)	32,16	32,16	32,16
Activation (FC)	ReLU	ReLU	ReLU
Output activation	Softmax	Softmax	Softmax
Loss function	Cross-entropy	Cross-entropy	Cross-entropy
Optimizer	Adam	Adam	Adam



(a) Pure DNN model



(b) 1D-CNN/DNN model



(c) LSTM/DNN model

Fig. 7 Architectures of the three neural network models evaluated in the experiment

Through this comprehensive approach, we aim to evaluate the effectiveness of different neural network architectures and temporal settings in improving the accuracy and robustness of wireless fingerprinting localization.

3.2 User tracking and association

This section describes the integration of wireless signals and images to associate the target users captured on the surveillance footage. First, we introduce the object detection model employed in our study. Then, a particle filter is applied to further track the users based on the localization results obtained from the fingerprinting model. Finally, we propose a user matching algorithm to match the detected object with the localization result to identify whether the person is the target user.

We adopted YOLOv7, the object detection model [20] to meet user recognition and performance requirements. YOLOv7 utilizes the re-parameterized technique, significantly reducing computational complexity and memory usage while improving inference speed. Additionally, it incorporates coarse-to-fine label assignments to enhance accuracy. This paper employs YOLOv7 to detect objects appearing in the scene, perform user tracking, and record obstacle information for subsequent user navigation.

The particle filter is a well-known method used in human tracking [21]. The key idea is to approximate the posterior probability in the Bayesian theorem based on the Monte Carlo sampling method. In other words, it uses particles to represent the system state and approximates the probability distribution of the target. Figure 8 shows the procedure of the particle filter that consists of the following steps:



Fig. 8 The flowchart illustrates the particle filter algorithm's main steps for user localization

- **Step 1. Initialization:** At the beginning, n particles are randomly generated and uniformly distributed on the map. Let $P = \{p_1, p_2, \dots, p_n\}$ be the set of particles, (x_i, y_i) represent the position of particle p_i , and w_i be the weight of p_i . At time $t = 0$, since the distribution is uniform, the weight of each particle is set to $\frac{1}{n}$.
- **Step 2. Update:** At $t = 1, 2, \dots, T$, we consider the estimated position result from the previous model as the prior position and use it to update the weights of all particles. First, we update the states of particles using (5). Δd represents the random movement, a Gaussian random variable with mean μ , and standard deviation σ is set to 1.4.

$$x_i = x_i + \Delta d_x, \quad y_i = y_i + \Delta d_y \quad (5)$$

Then, we update the weights of particles using (6). Let the prior position be denoted as p . We then calculate the likelihood between p and p_i . We multiply w_i by the likelihood and perform normalization to obtain $\hat{w}_i$ as the new weight. The closer p_i is to p , the higher the weight; conversely, the farther it is, the lower the weight.

$$w_i = w_i \cdot e^{\frac{-(p_i - p)^2}{2}}, \quad \hat{w}_i = \frac{w_i}{\sum_{i \in P} w_i} \quad (6)$$

- **Step 3. Resample:** Based on the results of Step 2, we perform resampling where particles with higher weights are selected more frequently, while particles with lower weights are selected less or even not chosen. This process helps adjust the distribution of particles to match the target's actual position better.
- **Step 4. Prediction:** The centroid of the particles is calculated based on (7) to obtain the estimated user location, where

$$\hat{p} = \sum_{i \in P} w_i \cdot p_i \quad (7)$$

3.3 User matching

To associate the localization result with the detected person, we propose a method to present and match the two outcomes on the display. We are interested in the user's situation within OZ, which is determined by the visible portion of the tactile paving in the image and serves as the range for collecting RP fingerprints during the offline phase. Then, OZ is divided into several observation blocks (OBs) to indicate an individual's location within OZ. In Fig. 9, $OB_i = RP_i$ indicates that the i -th OB corresponds to the i -th RP. Depending on the requirements, it is also possible for a single OB to correspond to multiple RPs.

The user matching algorithm is depicted in Fig. 10. Regarding the visual information, the bottom coordinate of the bounding box is used to determine which OB the person belongs

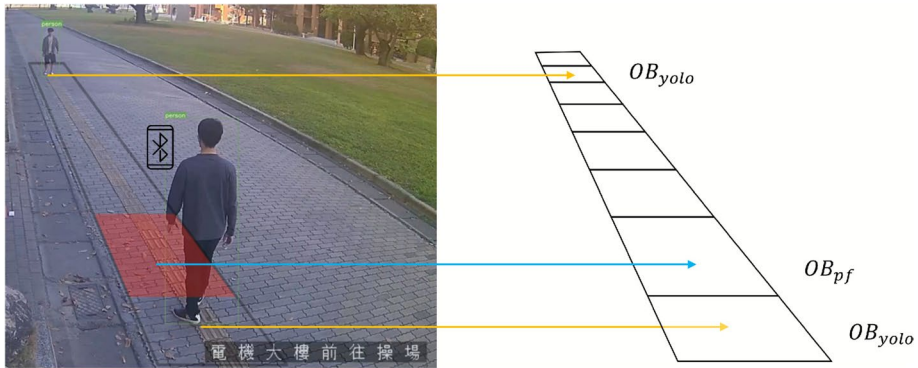


Fig. 9 Illustration of user matching on the tactile paving. The left panel shows a visually impaired user walking along the tactile path, with the wireless-predicted position (OB_{pf}) and visually detected positions (OB_{yolo}) indicated. The right panel schematically represents the alignment of wireless and visual observation blocks (OB_{pf} and OB_{yolo}) used for user matching in the ViTrack system

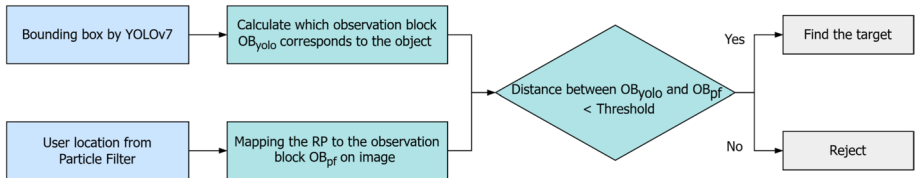


Fig. 10 The flowchart illustrates the user matching procedure in the ViTrack system, which combines YOLOv7's detected object's bounding box with the user location from the particle filter. A match is confirmed if the distance between observation blocks (OB_{yolo} and OB_{pf}) is below a threshold; otherwise, the candidate is rejected

to, denoted as OB_{yolo} . Regarding the wireless signal, if the localization result is RP_k , it is mapped to OB_k in the image, denoted as OB_{pf} . The determination is based on whether the difference between OB_{yolo} and OB_{pf} is less than the threshold. If it is less than the threshold, it is considered a successful match, indicating that the target user has been found; otherwise, it means the detection of a non-target pedestrian, and the candidate is rejected.

Our study sets the threshold to 2, considering the potential fluctuations in wireless signal-based localization. Figure 9 shows an example of user matching. In this image, two individuals are located within OZ, with the person closer to the camera carrying a mobile phone. The localization result of BLE fingerprinting is presented in red. The difference between OB_{yolo} and OB_{pf} exceeds the threshold for the person farther from the camera. Therefore, that person is considered a regular pedestrian. On the other hand, for the person closer to the camera, despite the BLE localization result being one block ahead of the person's actual position, the system determines that the person is a visually impaired user. We focus on the situation when users enter the OZ. Once a matching is completed, the subsequent localization is based on the image results. The user's coordinates in the image and information, such as the length and width of the bounding box and the BLE Universally Unique Identifier (UUID), will be recorded for subsequent user navigation and notification purposes.

4 Visual-based guiding system

In Fig. 6, the guiding process of ViTrack encompasses three core components: (1) body orientation, (2) deviation detection, and (3) obstacle avoidance. Figure 11 demonstrates the scenarios that will trigger ViTrack to send notifications to the visually impaired user regarding detected walking direction deviation and obstacle ahead alert. The user should follow the guiding direction in Fig. 11a. However, the body movement deviates in another perilous direction, potentially leading the user out of the safety zone or into a hazardous situation. In Fig. 11b, an obstacle in the form of a parked bicycle is observed within the safety zone, posing a potential collision risk as the user walks forward. The detection procedure will be discussed in detail in the following sections.

4.1 Human body orientation estimation

Human body orientation estimation (HBOE) is critical in navigation systems that provide real-time heading information and an individual's directional movements or turns. This capability is particularly significant for visually impaired users navigating unfamiliar environments. In the ViTrack system, camera-based tracking is used to monitor user movement and ensure adherence to a predefined virtual tactile pathway, referred to as the observation zone. This pathway is defined by tactile paving locations, as illustrated in Fig. 11. Unlike conventional navigation systems that rely on visual features such as colored lines or surface textures, ViTrack utilizes preconfigured tactile paving coordinates as navigation reference points, enhancing robustness to variations in lighting and surface conditions.

The HBOE module incorporates the MEBOW model [22] to estimate a user's body orientation with fine-grained precision. Upon detecting a user, the system extracts the RGB image region within the user's bounding box and inputs it into the HBOE model. The model produces a probability distribution over 72 discrete orientation bins, indexed from 0 to 71, with each bin representing a 5° segment of the full 360° circle. The bin with the highest probability is selected, and the corresponding body orientation angle θ_u is calculated by multiplying the bin index by 5° . The orientation angle increases counterclockwise, where



(a) Deviation



(b) Obstruction

Fig. 11 Example ViTrack notifications: **(a)** Deviation: User's direction diverges from the tactile path. **(b)** Obstruction: An obstacle (e.g., bicycle) on the tactile paving triggers an alert

0° corresponds to the user facing directly away from the camera (i.e., their back is visible), and 180° corresponds to the user facing directly toward the camera. The use of 5° resolution balances angular precision and computational efficiency, making it well-suited for real-time applications. Accurate body orientation estimation enables the system to infer the user's moving direction. In parallel, the system defines a preconfigured guiding direction, representing the desired heading or route. As shown in Fig. 11a, the system can assess whether the user is aligned with the intended path or requires directional correction by comparing the estimated moving direction with the guiding direction.

At its current development stage, ViTrack focuses on detecting user heading deviations relative to a predefined guiding direction and generating discrete navigational decisions: turn left, turn right, or keep straight. These decisions are derived from real-time comparisons between the user's estimated body orientation and the intended pathway. When a significant angular deviation is detected, the system classifies the necessary adjustment and triggers a corresponding navigational alert. This decision-making process forms the foundation of ViTrack's intelligent guidance capabilities. Once a directional decision is made, converting it into audio feedback using text templates, large language models (LLMs), or text-to-speech (TTS) systems is straightforward. For example, a "turn right" decision can be rendered as "Please turn right" or "Turn right ahead to stay on track." This modular design decouples orientation logic from the user interface, enabling flexible, context-aware audio feedback. The system operates over a test field modeled after real-world tactile pathways, allowing users to receive real-time movement correction alerts while navigating the observation zone. This implementation is key to ViTrack's broader mission of enhancing indoor navigation and accessibility, particularly for visually impaired individuals.

4.2 Deviation detection and correction framework

The proposed navigation framework ensures safe user guidance through hierarchical analysis of spatial boundaries and orientation patterns. Let $P_i \in \mathbb{R}^2$ denote the user's position in the camera coordinate system, monitored relative to two nested regions: the tactile path area R_{TP} (primary guidance zone) and its surrounding observation zone R_{OZ} (safety buffer), where $R_{TP} \subset R_{OZ}$. Perspective-aware calibration incorporates $\theta_{TP} \in (0^\circ, 180^\circ]$ representing the tactile path's orientation relative to each camera viewpoint (Fig. 12), with θ_u denoting the real-time user orientation estimated through body modeling.

Deviation detection follows three sequential checks (Algorithm 1). Critical deviations ($P_i \notin R_{OZ}$) trigger immediate alerts via $\text{Deviation} \leftarrow \text{TRUE}$. For users in the safety margin ($P_i \in R_{OZ} \setminus R_{TP}$), orientation compliance is evaluated using the acceptable interval:

$$I_A = \begin{cases} (\theta_{TP}, \theta_{TP} + 180^\circ), & \text{right TP section} \\ (\theta_{TP} + 180^\circ, \theta_{TP}), & \text{other sections} \end{cases} \quad (8)$$

where $\theta_u \notin I_A$ indicates potential drift. Normal movement ($P_i \in R_{TP}$) requires no intervention. For deviations, correction parameters derive from the rejection interval:

$$I_R = I_A^c = \begin{cases} (\theta_{TP} + 180^\circ, \theta_{TP}), & \text{right TP section} \\ (\theta_{TP}, \theta_{TP} + 180^\circ), & \text{other sections} \end{cases} \quad (9)$$

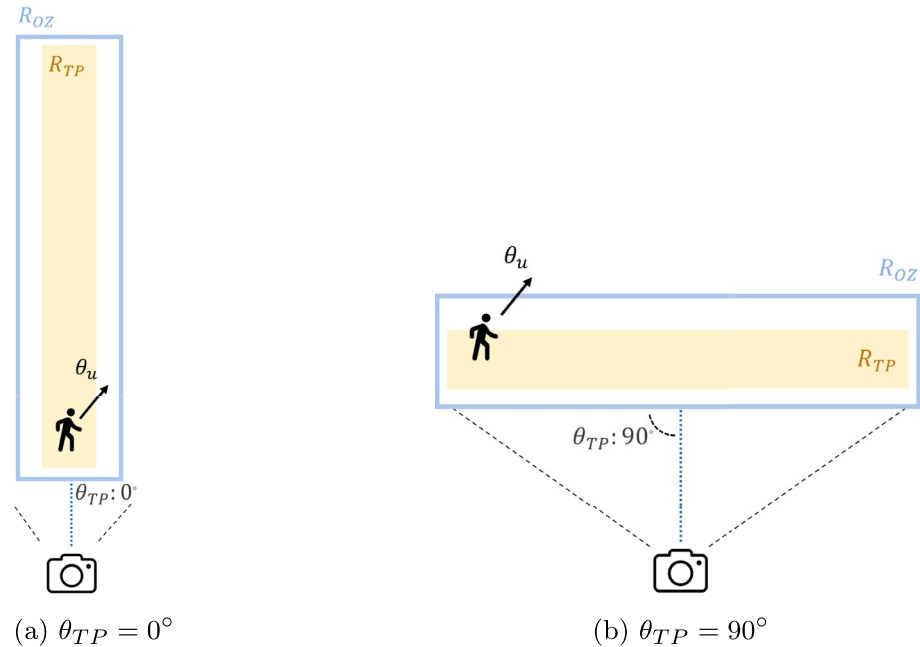


Fig. 12 Examples of configured θ_{TP} values in ViTrack, defined based on the camera's viewpoint relative to the tactile pathway

The system bisects I_R at midpoint $\theta_m = (\min(I_R) + \max(I_R))/2$, creating directional correction zones:

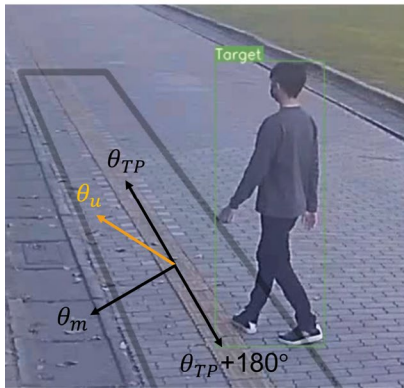
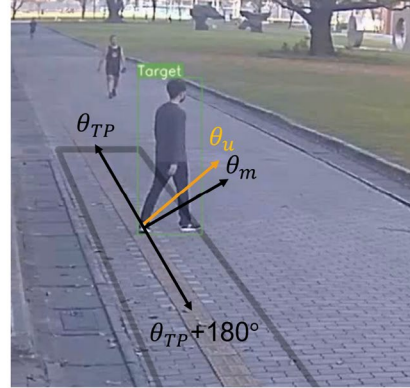
$$I_{r,1} = \begin{cases} (\theta_{TP} + 180^\circ, \theta_m), & \text{right TP section} \\ (\theta_m, \theta_{TP} + 180^\circ), & \text{other sections} \end{cases} \quad (10)$$

$$I_{r,2} = \begin{cases} (\theta_m, \theta_{TP} + 180^\circ), & \text{right TP section} \\ (\theta_{TP} + 180^\circ, \theta_m), & \text{other sections} \end{cases} \quad (11)$$

Guidance parameter N maps these intervals to navigation commands (Fig. 13):

$$N = \begin{cases} -1, & \text{advises moving to the right} \\ 1, & \text{advises moving to the left} \\ 0, & \text{no adjustment} \end{cases} \quad (12)$$

The framework achieves adaptive safety through three mechanisms: graduated intervention via nested regions $R_{TP} \subset R_{OZ}$, perspective-specific θ_{TP} calibration for varied camera angles, and $\mathcal{O}(1)$ computational complexity through spatial checks with interval arithmetic. Implementation requires perspective normalization matrices for spatial-angular relationships, with future enhancements including reinforcement learning for personalized θ_a adaptation, temporal convolutional networks for I_A interval prediction from path topology, and Inertial Measurement Unit (IMU) assisted orientation refinement. Continuous trajectory

(a) Acceptable θ_u direction(b) Reject θ_u direction: ViTrack warns the user should walk to the left.**Fig. 13** Determining deviation situations, where the user is at the right section of the TP

analysis (Fig. 13) shows $\theta_u \in I_A$ naturally converges toward R_{TP} , while $\theta_u \notin I_A$ triggers alerts through progressive divergence detection. Algorithm 1 assesses the N values, where $I_{r,1}$ activates rightward prompts ($N = -1$) and $I_{r,2}$ engages leftward guidance ($N = 1$), maintaining real-time operation on resource-constrained edge devices.

Algorithm 1 Deviation detection algorithm.

Input: Position P_i , zones R_{OZ}/R_{TP} , orientations θ_{TP}/θ_u
Output: Correction direction N , angle θ_a

```

1 if  $P_i \notin R_{OZ}$  then
2   |  $Deviation \leftarrow \text{TRUE}$ ;
3 else if  $P_i \in R_{OZ} \setminus R_{TP}$  then
4   | Calculate  $I_A$  via (8);
5   |  $Deviation \leftarrow \theta_u \notin I_A$ ;
6 else
7   |  $Deviation \leftarrow \text{FALSE}$ ;
8 end
9 if  $Deviation$  then
10  | Compute  $I_R$  using (9);
11  |  $\theta_m \leftarrow \frac{\min(I_R) + \max(I_R)}{2}$ ;
12  | if  $\theta_u \in (\min(I_R), \theta_m)$  then
13  |   |  $(N, \theta_a) \leftarrow (-1, |\theta_u - \min(I_R)|)$ ;
14  |   else if  $\theta_u \in (\theta_m, \max(I_R))$  then
15  |     |  $(N, \theta_a) \leftarrow (1, |\theta_u - \max(I_R)|)$ ;
16 else
17  |  $(N, \theta_a) \leftarrow (0, 0)$ ;
18 end

```

4.3 Obstacle avoidance method

The ViTrack system enhances navigation safety through real-time obstacle avoidance integrated with hierarchical behavioral orientation estimation (HBOE). As detailed in Algorithm 2, the framework processes obstacle data from YOLOv7 to adjust guidance commands (N') while generating collision warnings ($\mathcal{O}$). The core detection logic employs an 8-sector polar grid (Fig. 14) where $D[0 : 7]$ represents obstacle presence in:

0 : Front	4 : Back
1 : Front-Left	5 : Right-Back
2 : Left	6 : Right
3 : Left-Back	7 : Front-Right

The judgment of the obstacle's relative position to the user involves three coordinate transformations. We first calculate and record the grid with obstacles on the ground by computing the differences in the x and y coordinates between the obstacle (x_o, y_o) and the user (x_i, y_i) as follows:

$$dy = -(y_o - y_i) \quad dx = x_o - x_i \quad (13)$$

Then, it calculates the relative angle between them in the polar coordinate system as:

$$\theta_{o,u}^{(p)} = \frac{180^\circ}{\pi} \text{atan2}(dy, dx) \bmod 360 \quad (14)$$

However, due to the coordinate system, an adjustment is required:

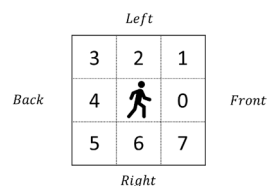
$$\theta_{o,u} = (\theta_{o,u}^{(p)} + \theta_{rotate} - \theta_u) \bmod 360 \quad (15)$$

This adjustment considers the specific coordinate system and provides $\theta_{o,u}$ to measure the obstacle's relative position to the user. θ_{rotate} represents the rotation angle for adjusting the coordinate system. In our study, $\theta_{rotate} = -90^\circ$ indicates that the system has been corrected by rotating counterclockwise by 90 degrees. Directional sector mapping uses angular intervals:

$$I_{dir}[j] = [45j - 22.5, 45j + 22.5) \bmod 360, \quad j \in \{0, \dots, 7\} \quad (16)$$

If $\theta_{o,u}$ is within $I_{dir}[j]$, then the presence of an obstacle in that direction is recorded. The algorithm also updates the deviation guidance detection N from Algorithm 1. The system

Fig. 14 Octagonal obstacle detection grid with sector numbering convention. Arrows indicate relative bearing calculation from user position P_i



checks for obstacles in the recommended walking direction and makes adjustments to the direction accordingly.

Algorithm 2 Obstacle-aware guidance adjustment.

Input: User position $\mathbf{P}_i$, guidance $N \in \{-1, 0, 1\}$ (left, none, right)
Output: Adapted guidance N' , obstacle flag $\mathcal{O}$

```

1 for  $i \leftarrow 0$  to 7 do
2    $D[i] \leftarrow 0$ 
3 end
4  $O_L \leftarrow \text{false}$ ,  $O_R \leftarrow \text{false}$ ,  $\mathcal{O} \leftarrow 0$ ;
5 foreach obstacle  $\mathbf{o}_k \in \text{YOLOv7}(\text{frame})$  do
6   if  $\|\mathbf{o}_k - \mathbf{P}_i\|_2 < r_{\text{safe}}$  then
7     Compute  $\theta_{o,u}$  via (13)-(15);
8     for  $j \leftarrow 0$  to 7 do
9       if  $\theta_{o,u} \in I_{\text{dir}}[j]$  (16) then
10         $D[j] \leftarrow 1$ ;
11         $\mathcal{O} \leftarrow 1$ ;
12      end
13    end
14 end
15 if  $D[1] \vee D[2]$  then
16   /* Left/front-left obstacles (Fig. 14 sectors 1-2) */
17    $O_L \leftarrow \text{true}$ ;
18 if  $D[6] \vee D[7]$  then
19   /* Right/front-right obstacles (Fig. 14 sectors 6-7) */
20    $O_R \leftarrow \text{true}$ ;
21 if  $(O_L \wedge (N = 1)) \vee (O_R \wedge (N = -1))$  then
22    $N' \leftarrow 0$  /* Cancel direction if obstacle in recommended
23   path */
24 else
25    $N' \leftarrow N$  /* Maintain original guidance */
26 end
27 return  $(N', \mathcal{O})$ ;

```

As shown in Fig. 15a, leftward correction ($N = 1$) is canceled when $O_L \leftarrow \text{true}$ detects left or front-left obstacles. Figure 15b demonstrates maintained course ($N' = 0$) with active warnings ($\mathcal{O} = 1$) for frontal obstacles. This dual-output design enables simultaneous path correction and collision prevention, which is critical for reliable navigation in campus environments.

4.4 Avoiding user mismatch through temporal consistency

User mismatch between wireless location estimates and visual detections can degrade the accuracy of deviation direction guidance, especially in the presence of multiple pedestrians. To address this, we propose a temporal consistency-based matching mechanism leveraging the continuity of user motion over time. Let the system operate over discrete time steps



(a) Obstacle on the left side

(b) Obstacle directly ahead

Fig. 15 Obstacle response scenarios: **(a)** Left correction canceled by a front-left obstacle ($D[1] = 1$ and $O_L = \text{true}$), **(b)** Frontal warning with maintained course ($\mathcal{O} = 1$)

$t_1, t_2, \dots, t_n$. At each step t_i , the wireless subsystem provides an estimated location P_i with an expected error radius d . We define the candidate set of visual detections within this radius as

$$\mathcal{C}_i = \{O_j \mid \|O_j - P_i\| \leq d, \quad j = 1, 2, \dots, M\}, \quad (17)$$

where O_j denotes detected objects in the camera view. Over the observation window of length n , each candidate object accumulates an appearance score based on the frequency of proximity to P_i :

$$S(O_j) = \sum_{i=1}^n \mathbb{I}(O_j \in \mathcal{C}_i), \quad (18)$$

where the indicator function $\mathbb{I}(\cdot)$ equals 1 if O_j is within radius d at time t_i , and 0 otherwise. The final user identity is selected as the object with the highest score:

$$O^* = \arg \max_{O_j} S(O_j). \quad (19)$$

This majority-vote approach reduces the impact of transient mismatches, as the probability of consistent error across n frames decreases approximately as

$$P_{\text{mismatch}} \approx \alpha^n, \quad (20)$$

where α is the per-frame mismatch probability. Hence, longer observation windows exponentially improve matching reliability. Choosing the observation window length n involves a trade-off between timely deviation detection and confidence in user matching, which we balance for real-time guidance. By ensuring robust long-term alignment between wireless

and visual data, this mechanism provides reliable user identification, which is critical for precise deviation direction guidance in multi-person scenarios (Fig. 16).

Algorithm 3 Temporal consistency-based user identification.

Input: Wireless estimates $\{P_i\}_{i=1}^n$, error radius d , visual detections $\{O_j\}_{j=1}^M$

Output: Chosen user O^*

```

1 for  $i \leftarrow 1$  to  $n$  do
2    $\mathcal{C}_i \leftarrow \{O_j \mid \|O_j - P_i\| \leq d, j = 1, \dots, M\}$ ;
3   foreach  $O_j \in \mathcal{C}_i$  do
4      $S(O_j) \leftarrow S(O_j) + 1$ ;
5   end
6 end
7  $O^* \leftarrow \arg \max_{O_j} S(O_j)$ ;
8 return  $O^*$ ;
```

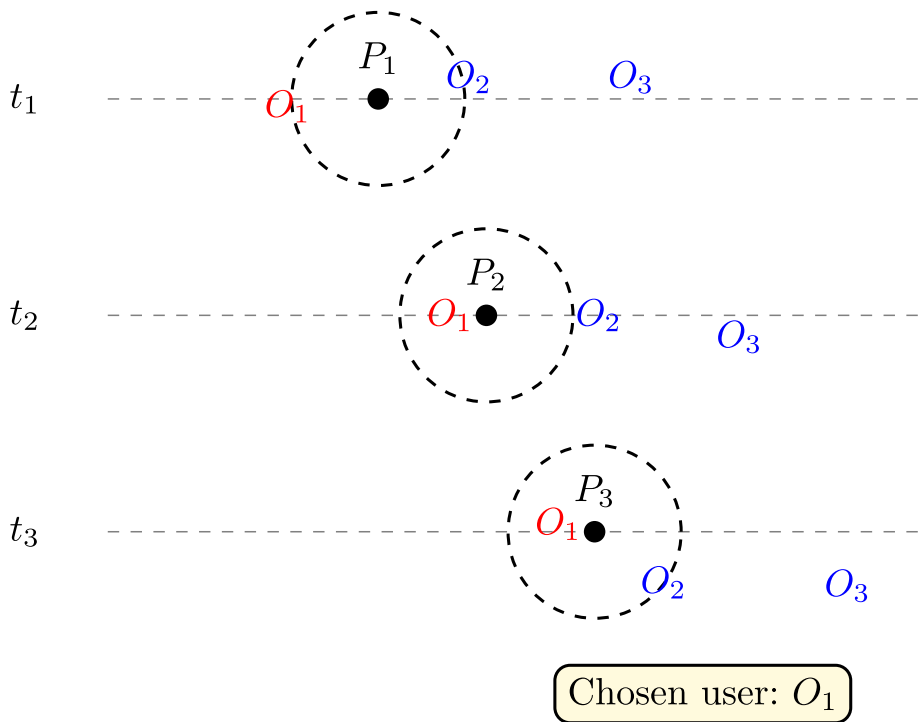
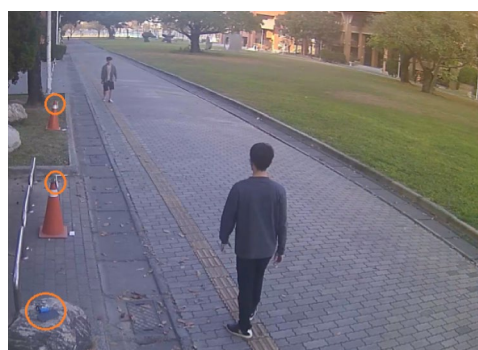


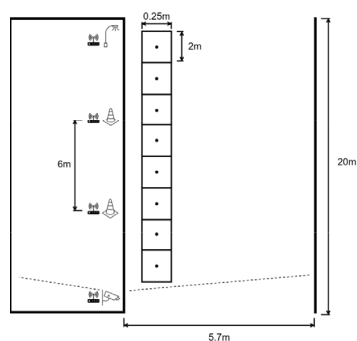
Fig. 16 Temporal consistency-based user identification scheme. At each time step t_1, t_2, t_3 , a circle centered at the wireless estimate P_i with error radius d is drawn. Candidate objects $\mathcal{C}_i$ are those $(O_1, O_2, \dots)$ falling inside the circle. Each object accumulates a score $S(O_j)$ equal to its number of appearances across the window. The final user is identified as $O^* = \arg \max_{O_j} S(O_j)$, which in this example corresponds to O_1

5 Experimental setup

The experiment starts when a user carrying a transmitting smartphone enters the OZ, triggering BLE detection. This initiates the user matching process, followed by the activation of visual-based guidance for the designated test routes. The experiment was conducted in the campus area outside the Department of Electrical Engineering building at NCKU in Taiwan. Figure 5 shows the camera that was used in the experiment. We selected this camera for its ability to capture a 16-meter length of tactile paving in a single image. As depicted in the floor plan in Fig. 17b, four wireless sniffers were strategically placed in the test field, with one camera outside its view. The dashed lines represent the perspective captured by the camera. The tactile paving section, i.e., the guiding area, is divided into eight subsections for location area management, each 2 meters long. These sub-sections have reference points (indicated by dot symbols) at their centers for positioning. Detailed system parameters can be found in Table 3.



(a) Sniffers



(b) The site measurement

Fig. 17 ViTrack evaluation test setup: (a) Outdoor tactile paving area with wireless sniffers marked in orange circles. (b) Site layout schematic showing dimensions ($20\text{m} \times 5.7\text{m}$), sniffer placement (6m apart), and tactile paving (0.25m wide, 2m between reference points)

Table 3 System parameters

Number of BLE Sniffers	4
Number of RPs	8
Camera vision area	$20\text{m} \times 5.7\text{m}$
Sniffer interval	6m
Bluetooth interface	version 5.0/ 2.4GHz
Advertising power	0 dBm
Advertising interval	100 ms
Training data collection time per RP	3 min

To assess the effectiveness of wireless-based user matching, we designed two testing routes shown in Fig. 18: “Back to the camera” and “Facing the camera.” A person traversed each route in five separate walks. In each instance, the user carried two different brands of mobile phones equipped with BLE signal transmitters, resulting in the collection of ten sets of test data. As the system identifies a *Guiding Request*, it initiates video streaming and consistently delivers precise directional guidance by detecting deviations and navigating around obstacles. To assess the effectiveness of a visual-based guiding system, we designed two testing routes shown in Fig. 19: “Deviation” and “Obstacle avoidance.”

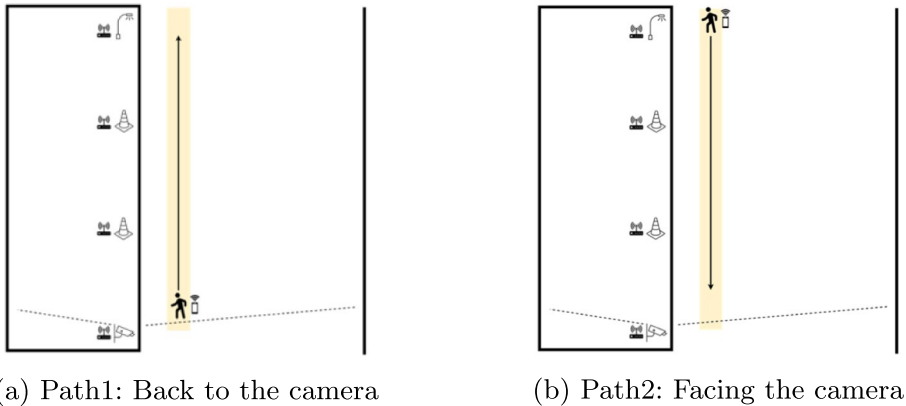


Fig. 18 Experimental scenarios for tracking accuracy evaluation: **(a)** Path 1: User walks along tactile paving with back to camera. **(b)** Path 2: User walks same path facing camera. Wireless sniffers are positioned along the route, highlighting the tactile paving used for evaluation

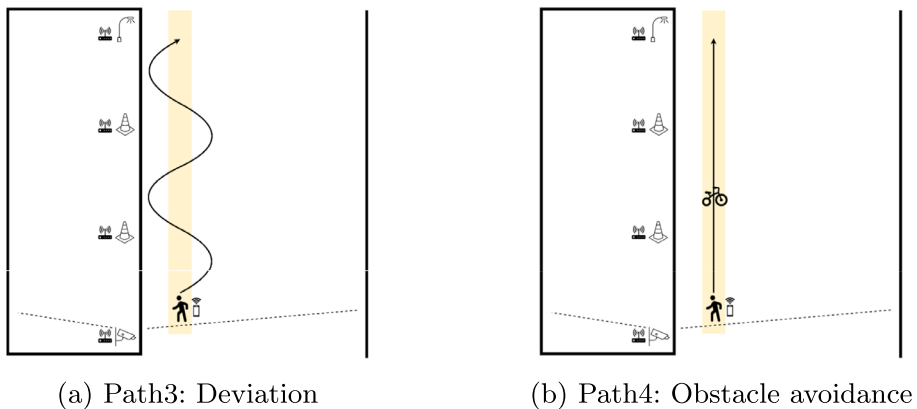


Fig. 19 Experimental scenarios for deviation detection and obstacle avoidance: **(a)** Path 3: User deviates from tactile paving, showing detection capability. **(b)** Path 4: User encounters an obstacle (bicycle), demonstrating the ViTrack system’s avoidance capabilities. Wireless sniffer positions are marked along the route

5.1 Training data collection

During the offline phase, we gathered wireless fingerprints at each Reference Point (RP) with participants carrying smartphones as BLE transmitters. This study employed smartphones from various brands, as depicted in Fig. 20a. The HTC U19e was utilized for collecting training data, while the Sharp AQUOS V was introduced in the testing phase to assess the system's performance across device heterogeneity. We utilized various smartphone brands to assess the robustness of our wireless user matching system. The HTC U19e was employed to gather training data, while the Sharp AQUOS V served as the testing device. This methodology simulates real-world device diversity and evaluates the system's capability to accommodate variations in BLE signal characteristics, ensuring reliable localization for users across different devices. We used the *nRF Connect* app to transmit BLE packets, providing the advantage of customizable transmission power and frequency during BLE advertising.

For the receiver, we employed the Raspberry Pi 3B+ as a BLE sniffer, depicted in both Fig. 20b and c. It is positioned at a height of 0.7 meters above the ground mounted on a traffic cone. These sniffers established connections with the ViTrack server through their wireless network adapter, consistently scanning BLE signals in the environment. If a received packet's UUID matched the service list, it was transmitted back to the server and recorded; otherwise, it was disregarded. The real-time video stream was acquired from the NCKU Surveillance System and is accessible through a streaming system. The image resolution is 1920×1080 , featuring a frame rate of 20fps.

During the online phase, we refresh the user's position using received wireless signals every second. In the event that a sniffer does not detect any wireless signals from the user within the specified interval, we utilize the RSSI value from the preceding second as the wireless fingerprint. This approach allows for the continuity of data collection in scenarios where real-time signal reception is interrupted. However, if a designated time threshold is surpassed, we set the RSSI value to 0, indicating that the user has moved beyond the observation range.

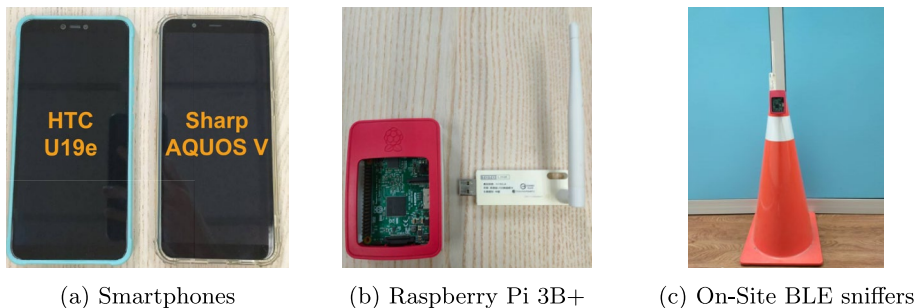


Fig. 20 Devices used in the experiments: (a) Smartphones (HTC U19e, Sharp AQUOS V) for BLE advertising and data collection; (b) Raspberry Pi 3B+ with Wi-Fi/BLE adapter for wireless sniffing; (c) BLE sniffer on a traffic cone for on-site deployment

6 Performance evaluation

To assess the tracking accuracy of the proposed system, we employ the mean distance error (MDE) as the performance metric. The MDE is calculated as the average Euclidean distance between the estimated location determined by the system and the corresponding ground truth location, representing the actual location. Here, (x, y) represents the coordinates of the estimated location, while $(\hat{x}, \hat{y})$ represents the coordinates of the ground truth location.

$$MDE = E[\sqrt{(x - \hat{x})^2 + (y - \hat{y})^2}] \quad (21)$$

In contrast, for the HBOE model utilized in the user guiding system, we use the absolute angle error as the performance metric. This metric illustrates the disparity in user body orientation between the predicted angle and the ground truth.

6.1 Performance on homogeneous and heterogeneous devices

First, we conduct a comparative analysis of the previously mentioned deep learning methods, assessing their performance on different devices. Since 1D-CNN and LSTM methods typically analyze data over specific time windows, we trained models using fingerprint features as inputs with time steps set to 1 and 3, respectively. Here, each time step corresponds to one set of RSSI feature vectors.

Figure 21 illustrates the MDE of four models across two distinct mobile device brands, namely U19e and AQUOS V. These devices possess diverse hardware configurations and exhibit variations in transmitted signal strengths among manufacturers. Notably, the U19e is utilized in the training phase, with the left part of the results in Fig. 21 representing the homogeneous test. Conversely, the right part of the results signifies the heterogeneous test.

It is evident that, in outdoor environments, the wireless fingerprinting method tends to exhibit lower effectiveness compared to indoor locations [21, 23], where it can typically achieve an error of less than one meter. However, due to the outdoor setting of our experiment, a higher MDE is inevitable, given the increased variability in RSSI values. It is

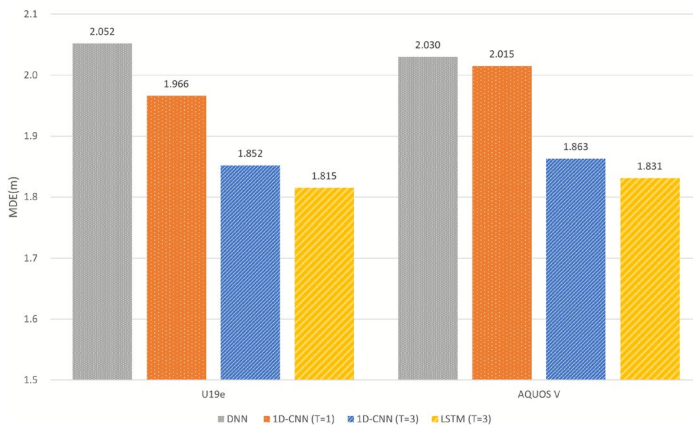


Fig. 21 MDE of four models on homogeneous and heterogeneous devices (Note that U19e is used to collect training data for the wireless fingerprint database)

important to note that in ViTrack, the primary objective of wireless localization is to identify the ViTrack user requesting assistance. In our experiment, each OB has dimensions of $2\text{ m} \times 2\text{ m}$, making an MDE less than 2 m sufficient for locating a smaller section of the OZ. If only one person is present in the area, identification is straightforward. However, with multiple people, ViTrack may take longer to capture the user. Once the user is captured, we leverage visual-based data to guide them.

Second, we observe that both testing cases exhibit similar performance. This implies that, in outdoor localization, the influence of hardware discrepancies on performance is relatively minor. Instead, the inherent variability of the wireless RSSI feature plays a more prominent role.

Third, we conducted a comparative analysis of four widely used methods for learning location based on RSSI data. Our findings reveal that both 1D-CNN ($T=3$) and LSTM ($T=3$) exhibit similar and notably superior performance compared to the other two schemes, namely DNN and 1D-CNN ($T=1$).

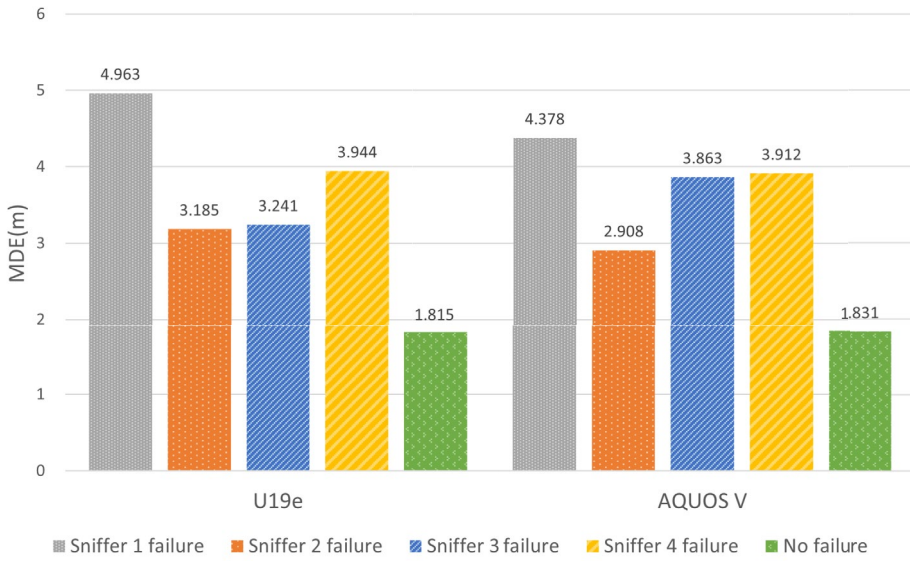
Specifically, with $T=1$, the MDE of the 1D-CNN model proves to be lower than that of the DNN model, underscoring the enhanced efficiency of the one-dimensional convolutional neural network in capturing local features, resulting in more precise localization. Notably, the 1D-CNN ($T=3$) model demonstrates a substantial improvement in accuracy compared to the 1D-CNN ($T=1$) model. This aligns with our expectations, as incorporating current and past signal information allows for leveraging a richer set of features, leading to enhanced accuracy in the localization process.

Finally, the integration of the DNN/LSTM model demonstrates superior performance. The synergistic use of a DNN in tandem with an LSTM proves beneficial for localization, leveraging the complementary strengths of both models. Particularly in tasks involving temporal sequences, such as signal strength variations over time or movements in a spatial context, the LSTM excels at effectively processing time-related aspects of the data. Concurrently, the DNN interprets intricate features extracted from raw inputs. This collaborative approach fosters a more comprehensive understanding of the data, resulting in improved performance in localization tasks compared to a standalone DNN or a DNN/CNN model.

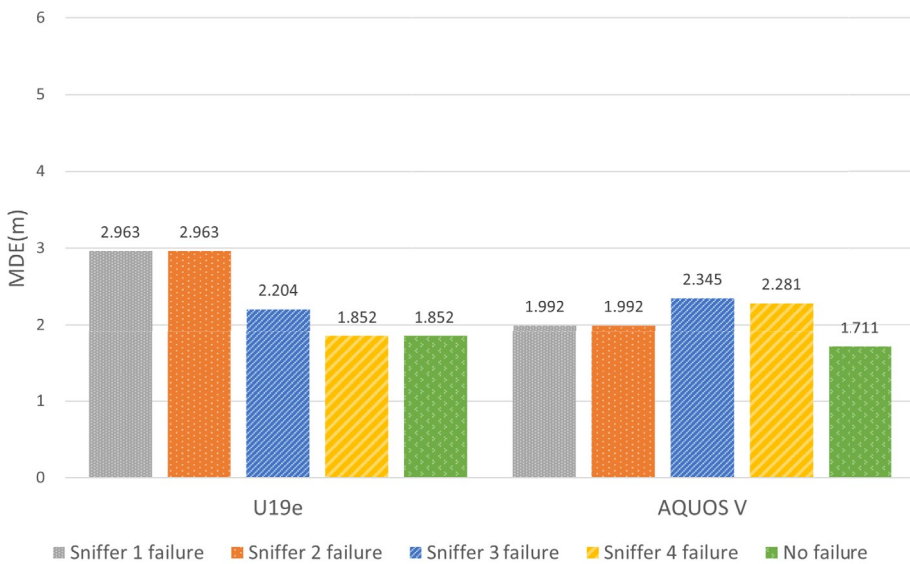
6.2 Wireless sniffer failure

This subsection evaluates system performance when confronted with wireless sniffer failures. Our experimental setup involves the deployment of four sniffers, denoted as Sniffers 1, 2, 3, and 4, strategically positioned over the OZ, as illustrated in Fig. 17b. Two distinct methods are compared for their efficacy. The baseline method relies solely on training with the original dataset collected during the training phase. In contrast, our enhanced method undergoes training with a hybrid dataset comprising both the original training data and augmented data. The augmented dataset comprises four independent groups, each containing labeled data. Notably, each group is tailored to simulate the failure of a specific sniffer, with that particular sniffer's data omitted from the training set. Both methods utilize the DNN/LSTM model with the lowest MDE from prior tests.

As depicted in Fig. 22a and b, our improved method consistently outperforms the baseline when facing sniffer failures. For the U19e testing case, when Sniffer 1 fails, MDE decreases from 4.963m to 2.963m, yielding a performance improvement of 40.3%. In the



(a) Baseline method: Training with original data



(b) Our improved method: Training with data augmentation

Fig. 22 Performance of two training methods against sniffer failure

case of Sniffer 4 failure, MDE decreases from 3.944m to 1.852m, indicating a 53.04% improvement. Similarly, when Sniffer 1 fails, MDE decreases from 4.378m to 1.992m for the AQUOS V testing case, showing a 54.5% performance gain. In the case of Sniffer 4 failure, MDE decreases from 3.912m to 2.281m, reflecting a 41.69% improvement. These results show the effectiveness of our improved training approach in reducing the MDE under sniffer failure. Our augmentation of the data enables the model to simulate a broader range of real-world scenarios, providing a more comprehensive learning experience and enhancing the overall robustness of the system.

6.3 Results of user body orientation estimation

In ViTrack, deviation detection and obstacle avoidance are based on the user's body orientation relative to the guiding path. Therefore, the precision of the HBOE model plays a pivotal role in ensuring effective user navigation. To assess the HBOE model's accuracy in ViTrack, we evaluated the absolute angle difference for two participants, A and B, as they traversed a designated route at different speeds, as illustrated in Fig. 19a. During the tests, conducted at varying walking speeds categorized as "fast" (0.8m/s to 1m/s) and "slow" (0.4m/s to 0.6m/s), a camera aligned with the tactile paving direction recorded ground truth data.

The results, depicted in Fig. 23, illustrate the absolute angle differences for both participants across different walking speeds. Participant A consistently exhibited a median angle difference of approximately 5 degrees under fast and slow walking conditions. Conversely, these median differences for participant B were 10 degrees and 5 degrees, respectively. These findings highlight ViTrack's ability to provide accurate body orientation estimation, accommodating users with diverse walking speeds and ensuring reliable navigation services.

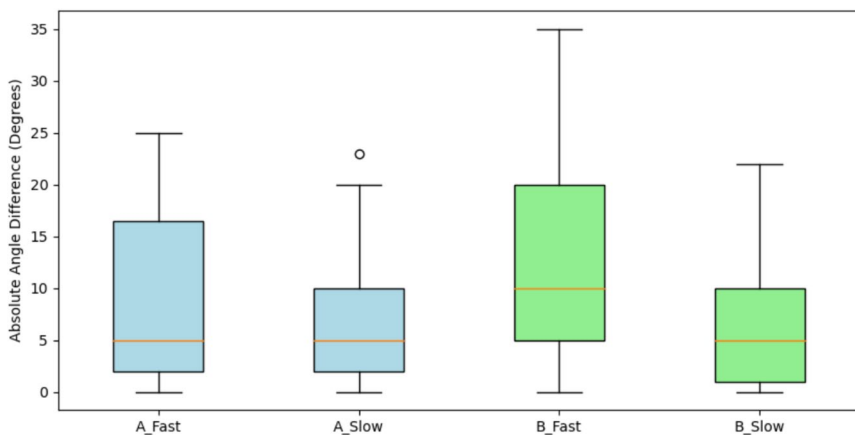


Fig. 23 Absolute angle differences for different walking speeds and participants

6.4 Overall system performance summary

The evaluations in this section validate the effectiveness of ViTrack's core components. Wireless localization demonstrates robust performance across homogeneous and heterogeneous devices, with the LSTM model outperforming other architectures and data augmentation mitigating sniffer failures. This confirms its suitability for initial user detection and coarse positioning within the OZ. Complementing this, the HBOE model achieves high accuracy in determining user heading, enabling precise deviation detection and obstacle avoidance. While BLE provides initial localization (1–3 m accuracy) for user identification, the vision-based system refines tracking to a minor error during guidance. These results collectively validate ViTrack's design: wireless modalities enable coarse user matching, while vision ensures accurate orientation detection and dynamic guidance within the camera's field of view.

7 Conclusion and future work

ViTrack presents a sustainable and user-centered navigation solution for visually impaired individuals by harnessing existing surveillance infrastructure and wireless technologies. The system eliminates the need for users to hold or interact with additional devices, prioritizing comfort, safety, and daily practicality, especially for those who rely on white canes. ViTrack delivers real-time, third-person camera-based guidance that reduces occlusion risks and enhances detection reliability compared to body-worn or high-maintenance wearable solutions by leveraging low-power, infrastructure-based sensing, and fixed cameras. This study illustrates how integrating BLE-based localization with vision-based orientation and obstacle detection can effectively address the shortcomings of traditional navigation aids. The architecture of the system facilitates automatic user identification, precise tracking through LSTM neural networks, and reliable guidance, even amidst device variability or sniffer malfunctions, owing to efficient data augmentation strategies. Collectively, these advancements reinforce ViTrack's design philosophy: to provide accessible, scalable, and cost-effective navigation support by utilizing existing resources and minimizing environmental impact. Key findings indicate that ViTrack successfully achieves dependable initial localization and user matching through BLE. Concurrently, vision-based techniques offer accurate orientation and dynamic obstacle avoidance within the camera's field of view. The system's real-time feedback effectively meets the immediate needs of visually impaired users, such as following paths and detecting hazards, without adding extra device or interaction demands.

Several key initiatives will enhance ViTrack's capabilities. A primary focus will be evaluating system performance with multiple simultaneous users and specifically assessing guidance reliability and false match rates in more dynamic and crowded environments. We aim to extend ViTrack's functionality to accommodate more complex scenarios by integrating multiple observation zones and utilizing movement prediction alongside IMU data for seamless navigation. Furthermore, we will investigate mounting options for smartphones that facilitate more stable IMU data collection without requiring users to hold the device physically. Future developments will also encompass predictive tracking of moving obstacles, coordination among multiple cameras, and trajectory modeling to support robust navi-

gation in larger, more dynamic spaces. For deployments in busier areas, enhancements such as NFC checkpoints at the entry points of observation zones and integration with advanced devices such as multi-band GPS smartwatches will be considered to improve positioning accuracy and overall user experience. By prioritizing sustainable design, user comfort, and the practical reuse of existing infrastructure, ViTrack emerges as a scalable and effective alternative to expensive, high-tech solutions. This initiative illustrates that the thoughtful integration of available resources can substantially benefit visually impaired individuals in their daily environments. As part of our future work, we plan to recruit visually impaired users to conduct real-world evaluations and further validate the usability and performance of ViTrack in daily navigation scenarios.

Appendix: Privacy, ethics, and GDPR compliance

Our approach to privacy and data protection is based on minimizing data collection, ensuring user autonomy, and strict adherence to relevant regulations. Future work will continue to follow these protocols and conduct further privacy and ethics reviews as we expand the scope of user studies.

Bluetooth data RSSI signals are collected solely when a participant enables Bluetooth and runs the navigation app. Data consists of timestamps, device IDs, and signal values, with no personally identifiable information.

Camera data Campus surveillance cameras are triggered only when a user requests guidance. Images are not stored; only anonymous object IDs are extracted for spatial matching. No faces, biometrics, or personal images are recorded or retained. During guidance, only the location and movement trajectory of the object are utilized.

Anonymization methods All camera data is processed in real time with only object IDs extracted. No raw video or facial features are stored. Data minimization and anonymization are employed throughout.

GDPR compliance Data handling follows GDPR principles:

- Lawfulness, fairness, and transparency: Data collection is fully user-controlled, and informed consent is obtained.
- Data minimization and purpose limitation: Only essential, anonymized data is collected.
- Security: Data is encrypted and accessible only by authorized research personnel.
- Voluntary participation: Users have complete autonomy to opt in or out of the system.

Ethical considerations All user studies are approved by the relevant ethical committee. Special precautions are taken to protect the rights and privacy of vulnerable populations, including visually impaired participants.

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Availability of supporting data This RSSI data used in this paper is based on our training dataset and testing dataset, which can be downloaded online, via <https://github.com/Shami0306/BLE/tree/main/>. This body movement data used is based on the MEBOW dataset, which can be downloaded online via <https://chenyanwu.github.io/MEBOW/>. Finally, our experimental data for the angle differences can be downloaded via http://github.com/Jerejere0808/deviation_angle.

Declarations

Ethical approval Not Applicable.

Competing interests Not Applicable.

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